

国防科技大学 2019—2020 学年春季学期

《概率论与数理统计》考试试卷 (B) 卷 参考解答

可能用到的常数: $u_{0.975}=1.960$, $\chi_{0.1}^2(19)=27.204$

一、填空题(本题每小题 3 分, 满分 30 分.)

1. 0.1

2. $24/5$

3. $f_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)} = \begin{cases} \frac{1}{2x}, & -x < y < x, \\ 0, & \text{其它 } y. \end{cases}$

4. $\frac{2}{3}e^{-2}$

5. .23.4

6. 1/2

7. $c = \sqrt{5}$, 自由度为 8 的 t 分布

8. $\hat{\mu}_2$

9. $(\bar{X} \pm \frac{\sigma}{\sqrt{n}} z_{\alpha/2}) = (580 \pm 9.8) = (570.2, 589.8)$

二、解答题(共 6 题, 70 分.)

10、(10 分) 解 设 A_i 表示“第 i 枚导弹捕获卫星”, B_i 表示“第 i 枚导弹捕获卫星后将其击毁”

C_i 表示“第 i 枚导弹击毁卫星”($i=1,2,3$), C_1, C_2, C_3 相互独立, 且

$$P(C_i) = P(A_i B_i) = P(A_i)P(B_i | A_i) = 0.7 \times 0.6 = 0.42 \quad (i=1,2,3) \dots\dots\dots (5 \text{ 分})$$

则实施该计划将失控卫星击毁的概率

$$\begin{aligned} p &= P(C_1 \cup \bar{C}_1 C_2 \cup \bar{C}_1 \bar{C}_2 C_3) \\ &= P(C_1) + P(\bar{C}_1 C_2) + P(\bar{C}_1 \bar{C}_2 C_3) \\ &= P(C_1) + P(\bar{C}_1)P(C_2) + P(\bar{C}_1)P(\bar{C}_2)P(C_3) \\ &= 0.42 + (1-0.42) \times 0.42 + (1-0.42)^2 \times 0.42 \\ &= 0.805 \end{aligned} \dots\dots\dots (10 \text{ 分})$$

11、(10 分) 解 令 $y = g(x) = \ln x, (x > 0)$, 其反函数为 $h(y) = e^y, h'(y) = e^y, (-\infty < y < \infty)$, 于是

$$f_Y(y) = |h'(y)|f(h(y)) = \frac{2e^y}{\pi(1+e^{2y})}, \quad (-\infty < y < \infty).$$

12、(10 分) 解: $f_Z(z) = \int_{-\infty}^{+\infty} f(x, z-x) dx = \int_0^1 f(x, z-x) dx, \quad \dots\dots 3 \text{ 分}$

当 $z \leq 0, \text{ or } z \geq 2$ 时, $f(x, z-x) = 0. \Rightarrow f_Z(z) = 0;$

当 $0 < z \leq 1$ 时,

$$f_Z(z) = \int_{z/2}^z 3x dx = \frac{9}{8} \cdot z^2; \quad \dots\dots 6 \text{ 分}$$

当 $1 < z < 2$ 时, $f_Z(z) = \int_{z/2}^1 3x dx = \frac{3}{2}(1 - \frac{z^2}{4}) \dots\dots 8 \text{ 分}$

$$\therefore f_Z(z) = \begin{cases} \frac{9}{8} \cdot z^2 & , \quad 0 < z \leq 1 \\ \frac{3}{2}(1 - \frac{z^2}{4}) & , \quad 1 < z < 2 \\ 0 & , \quad \text{其它} \end{cases} \quad \dots\dots 10 \text{ 分}$$

13、解 $E(X) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xf(x, y) dx dy = \int_0^1 \int_0^1 x(2-x-y) dx dy = \frac{5}{12}$

$$\therefore E(X^2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x^2 f(x, y) dx dy = \int_0^1 \int_0^1 x^2(2-x-y) dx dy = \frac{3}{12}$$

$$\therefore D(X) = E(X^2) - [E(X)]^2 = \frac{3}{12} - (\frac{5}{12})^2 = \frac{11}{144}$$

$$E(XY) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xyf(x, y) dx dy = \int_0^1 \int_0^1 xy(2-x-y) dx dy = \frac{1}{6}$$

由对称性有 $E(Y) = E(X) = \frac{5}{12}, D(Y) = D(X) = \frac{11}{144}$, 故

$$\text{cov}(X, Y) = E(XY) - E(X)E(Y) = \frac{1}{6} - \frac{5}{12} \cdot \frac{5}{12} = -\frac{1}{144}$$

14、(15 分) 解: 1) 似然函数为: $L(\theta) = \prod_{i=1}^n f(x_i) = \prod_{i=1}^n \frac{2x_i}{\theta} e^{-x_i^2/\theta} = \frac{\prod_{i=1}^n 2x_i}{\theta^n} e^{-\frac{1}{\theta} \sum_{i=1}^n x_i^2}$ 3 分

对数似然函数为: $l(\theta) = \ln L(\theta) = \sum_{i=1}^n \ln 2x_i - n \ln \theta - \frac{1}{\theta} \sum_{i=1}^n x_i^2$ 5 分

$$\frac{d}{d\theta} l(\theta) = -\frac{n}{\theta} + \frac{1}{\theta^2} \sum_{i=1}^n x_i^2 = 0 \Rightarrow \hat{\theta} = \frac{1}{n} \sum_{i=1}^n x_i^2 \quad \text{.....10 分}$$

2) $E(X^2) = \int_{-\infty}^{\infty} x^2 f(x) dx = \int_0^{\infty} x^2 \frac{2x}{\theta} e^{-x^2/\theta} dx = \int_0^{\infty} x^2 \frac{1}{\theta} e^{-x^2/\theta} dx^2 = \int_0^{\infty} t \frac{1}{\theta} e^{-t/\theta} dt = \theta$

所以是无偏估计。15 分

15、(15 分) 解 依题意, 要求检验

$$H_0: \sigma^2 \leq \sigma_0^2 = 450, \quad H_1: \sigma^2 > \sigma_0^2$$

采用单边 χ^2 检验法, 求得 H_0 的拒绝域为

$$\frac{(n-1)S^2}{\sigma_0^2} > \chi_{\alpha}^2(n-1) = \chi_{0.1}^2(19) = 27.204$$

由题给数据算得

$$\frac{(n-1)S^2}{\sigma_0^2} = \frac{19 \times 663.2}{450} = 28.002 > 27.204$$

故拒绝 H_0 , 即可以认为这批产品的方差不符合要求。